Climate Informatics 2026 (2026), xx: 1–3
doi:10.1017/xxxx

CAMBRIDGE UNIVERSITY PRESS

CLIMATE INFORMATICS 2026
EXTENDED ABSTRACT

Neural-ROM: A Graph Neural Network for Probabilistic Regional Ocean Forecasting

Daniel Holmberg[1], Joel Oskarsson[2*], Erik Larsson[3], Fredrik Lindsten[3] and Teemu Roos[1]

[1]University of Helsinki, Helsinki, Finland
[2]ETH AI Center, ETH Zurich, Zurich, Switzerland
[3]Linköping University, Linköping, Sweden
*Corresponding author. Email: joel.oskarsson@outlook.com

Keywords: Oceanography, Ensemble forecasting, Generative models, Probabilistic modeling

1. Introduction

Accurate and timely ocean forecasts are crucial for maritime activities, including shipping, aquaculture, and environmental management. Numerical oceanographic models are today computationally intensive, limiting the number of forecasts that can be produced in operational settings. In particular this hinders running large ensemble forecasts in order to quantify uncertainty in estimates of future ocean states. Recent advances in data-driven oceanography have demonstrated forecasting skill surpassing that of operational numerical models, both in kilometer-scale regional settings (Holmberg et al., 2025) and in global eddy-resolving settings (Cui et al., 2025). However, these models remain deterministic, leaving uncertainty quantification through ensemble ocean forecasting an open challenge. By taking inspiration from similar challenges in weather prediction (Lang et al., 2024; Oskarsson et al., 2024) we introduce Neural-ROM, a probabilistic machine learning Regional Ocean Model (ROM). Combining a Graph Neural Network (GNN) architecture and a probabilistic latent variable formulation, Neural-ROM is a generative machine learning model capable of accurate ocean forecasting and efficient ensemble generation.

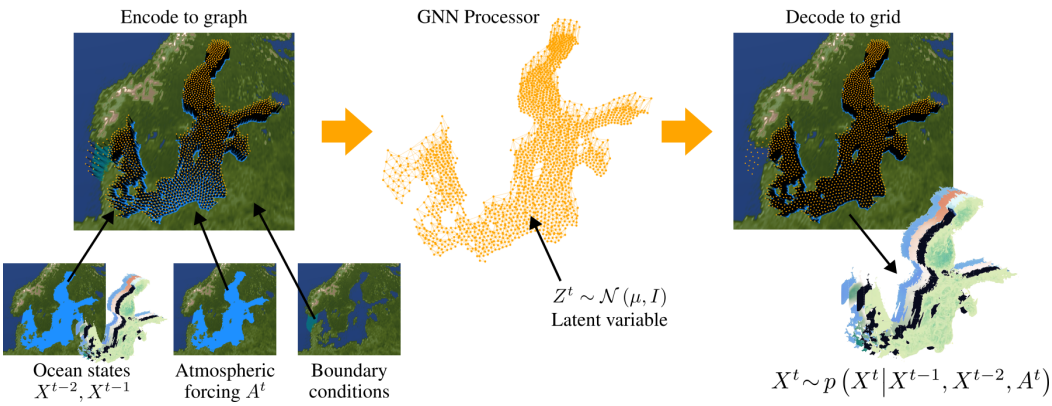

Figure 1. *Overview of sampling a one step prediction of X^t with the Neural-ROM model. Longer forecasts are rolled out by feeding sampled predictions back to the model at the following time steps.*

© The Authors(s) 2020. Published by Cambridge University Press. This is an Open Access article, distributed under the terms of the Creative Commons Attribution licence (http://creativecommons.org/licenses/by/4.0/), which permits unrestricted re-use, distribution, and reproduction in any medium, provided the original work is properly cited.

2. Methods and data

We are here interested in a probabilistic model for ocean dynamics and biogeochemistry. Specifically, given gridded ocean states X^{t-1}, X^{t-2} from previous time points and atmospheric forcing A^t we aim to model the distribution $p\left(X^t \middle| X^{t-1}, X^{t-2}, A^t\right)$ over ocean states at the next time step. We model this distribution as a deep generative model, leveraging a latent random variable Z^t and a GNN f_θ to produce samples. To generate X^t we first draw a Gaussian sample $Z^t \sim \mathcal{N}(\mu, I)$, and pass this through the neural network together with conditioning information in order to produce a sample $X^t = f_\theta\left(Z^t, X^{t-1}, X^{t-2}, A^t\right)$.

The Neural-ROM architecture is inspired by the Graph-EFM weather model (Oskarsson et al., 2024) and builds on a hierarchical graph-based approach. In f_θ the gridded inputs are mapped to a coarser mesh graph and spatial interactions modeled through graph neural network layers. As shown in fig. 1, this type of model architecture effectively handles the complex, irregular geometry of ocean grids. While our first approach uses a regular layout for the mesh graph, we also experiment with a novel graph-generation technique based on spatial clustering. This allows for better matching the graph geometry to bays and islands. Additionally, it enables model inputs on different grids, allowing for keeping the atmospheric forcing on its original coarser grid.

Neural-ROM is trained on a time series of ocean reanalysis data, matching the model distribution with the true distribution of the data. As training objectives we rely on a combination of the Evidence Lower Bound (ELBO) (Oskarsson et al., 2024) and direct optimization of the Continuous Ranked Probability Score (CRPS) (Lang et al., 2024).

We evaluate Neural-ROM on a regional forecasting problem, producing ensemble forecasts for the Baltic sea. In this regional setting the model takes an additional input in the form of lateral boundary conditions at the Skagerrak strait, which ensures consistency with the global ocean. We train the model on Baltic Sea data from the Nemo-Nordic reanalysis and WAM hindcast (E.U. CMEMS, 2024), at one nautical mile resolution. Both surface variables and multiple depth levels are modeled, encompassing ocean physics (currents, temperature, salinity, sea surface height) biogeochemistry (chlorophyll, oxygen), and key wave parameters, all on a daily basis. Atmospheric forcing during training comes from the ERA5 reanalysis.

3. Preliminary results

Initial results show that our model is capable of producing useful and physically plausible forecasts up to multiple weeks ahead. We compare Neural-ROM to SeaCast (Holmberg et al., 2025), a deterministic GNN-based model, and find that our ensemble mean yields significantly lower errors at longer lead times. Importantly, the ensemble exhibits diversity between sampled forecasts, representing uncertainty in hard to forecast quantities. While some work remains to ensure calibrated ensemble spread for all variables, these results indicate great potential in the approach for probabilistic ocean prediction. In the continuation of the project we aim to further experiment with the graph generation methodology as well as extending our experiments to additional domains.

Acknowledgments. This research was supported by the ETH AI Center through an ETH AI Center postdoctoral fellowship to Joel Oskarsson, the Research Council of Finland (grant no: 361902), the Swedish Research Council (grant no: 2020-04122, 2024-05011), the Wallenberg AI, Autonomous Systems and Software Program (WASP) funded by the Knut and Alice Wallenberg Foundation, and the Excellence Center at Linköping–Lund in Information Technology (ELLIIT). Our computations were enabled by the LUMI supercomputer, owned by the EuroHPC Joint Undertaking and hosted by CSC–IT Center for Science, and the Berzelius resource at the National Supercomputer Centre, provided by the Knut and Alice Wallenberg Foundation.

References

Cui, Y., Wu, R., Zhang, X., Zhu, Z., et al. (2025). Forecasting the eddying ocean with a deep neural network. *Nature Communications*.

E.U. CMEMS (2024). Baltic sea physics reanalysis, Baltic sea biogeochemistry reanalysis and Baltic sea wave hindcast.

Holmberg, D., Clementi, E., Epicoco, I., and Roos, T. (2025). Accurate mediterranean sea forecasting via graph-based deep learning. *Scientific Reports*.

Lang, S., Alexe, M., Clare, M. C., Roberts, C., et al. (2024). AIFS-CRPS: ensemble forecasting using a model trained with a loss function based on the continuous ranked probability score. *arXiv preprint arXiv:2412.15832*.

Oskarsson, J., Landelius, T., Deisenroth, M. P., and Lindsten, F. (2024). Probabilistic weather forecasting with hierarchical graph neural networks. In *Advances in Neural Information Processing Systems*, volume 37.